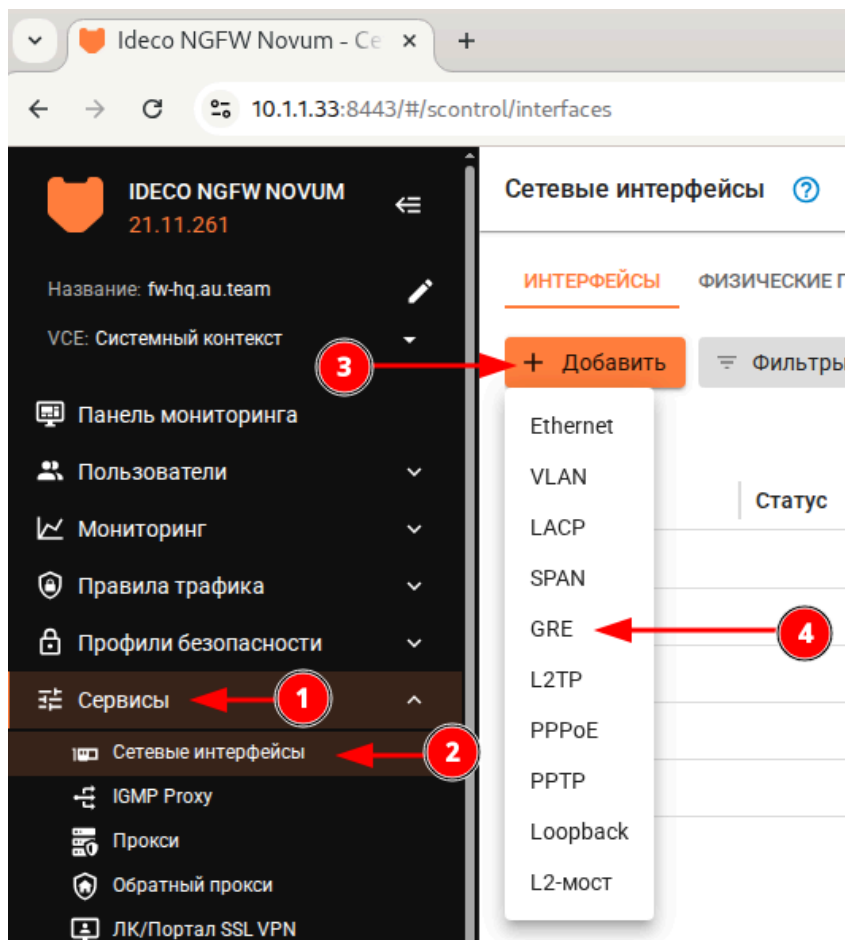


ADM-HQ:

- Создаём туннельный интерфейс `tunnel.1` в сторону **RTR-BR** с IP-адресом `10.0.1.1/30`:



- заполняем форму **Добавление GRE-туннеля**

Добавление GRE-туннеля

Название

tunnel.1

Настройки

Роль

WAN

Зона

Поле необязательное

VCE

Системный контекст

Виртуальный контекст (VCE), в котором будет использоваться интерфейс

Интерфейс

ISP

Локальный IP-адрес интерфейса

63.27.18.18

Поле необязательное. Если IP-адрес не задан вручную, автоматически назначается один из IP-адресов интерфейса.

Адрес удалённого интерфейса

84.212.78.78

IP-конфигурация

IP-адрес интерфейса туннеля

10.0.1.1/30

+ Добавить IP-адрес с маской

- Должно получиться следующее:



+ Добавить

Фильтры

Отображение

Поиск

Тип	Статус	Настройки		Зона	VCE	Интерфейс/порт	Ter VLAN	IP-конфигурация
		Название ↑	Роль					IP-адрес/маска
Ethernet		ISP	WAN	—	● Системный ко...	eth0_f1	—	63.27.18.18/23
Ethernet		mgmt	LAN	—	● Системный ко...	eth1_cb	—	—
GRE		tunnel.1	WAN	—	● Системный ко...	ISP	—	10.0.1.1/30
VLAN		vlan10	LAN	—	● Системный ко...	mgmt	10	10.1.1.1/27
VLAN		vlan20	LAN	—	● Системный ко...	mgmt	20	10.1.1.33/28
VLAN		vlan30	LAN	—	● Системный ко...	mgmt	30	10.1.2.1/24

- Аналогично и для tunnel.2:
 - ожидаемый результат

Добавление маршрута

Название ← 1

Поле необязательное

Назначение ← 2

Все Выбраны

Поиск

✓ ☐ Специальные · 1

✓ ☐ IP-адреса · 1

+ Добавить объект ← 3

1

Добавление объекта

Тип ←

Название ←

Значение ←

Комментарий

0/256

Добавить Отмена

Добавление объекта

Тип ←

Название ←

Значение ←

Комментарий

0/256

Добавить Отмена

Добавление объекта

Тип **Подсеть**

Название **RTR-BR <-> FW-BR**

Значение **10.2.0.0/30**

Комментарий

0/256

Добавить **Отмена**

Статическая ?

Добавление маршрута

Название **BR**

Поле необязательное

Назначение **IP BR vlan10 +2**

Интерфейс **tunnel.1**

Шлюз

▼ Пользователи · 15

▼ IP-адреса · 1

+ Добавить объект

+ Добавить значение

Добавление объекта

Тип **IP-адрес**

Название **RTR-BR tunnel.1**

Заполняется автоматически из значения, если оставить пустым

Значение **10.0.1.2**

Комментарий

0/256

Добавить **Отмена**

Добавление маршрута

Название
BR

Поле необязательное

Назначение
IP BR vlan10 X + 2 X

Интерфейс
tunnel.1 X

Шлюз
RTR-BR tunnel.1 X

Поле необязательное. Укажите шлюз, если выбран интерфейс с режимом статической IP-конфигурации.

Приоритет
10

Введите число от 1 до 255. Чем меньше приоритет, тем приоритетнее маршрут

Дополнительно

Комментарий
0/256

Добавить

Отмена

- Результат:

+ Добавить Фильтры Отображение Поиск

Название	Назначение	Интерфейс	Шлюз	Приоритет	Статус
BR	IP BR vlan10 IP BR vlan20 IP RTR-BR <-> FW-BR	tunnel.1	IP 10.0.1.2	10	Используется

- Добавляем аналогичным образом маршрут в сеть COD:
 - ожидаемый результат

+ Добавить Фильтры Отображение Поиск

Название	Назначение	Интерфейс	Шлюз	Приоритет	Статус
BR	IP BR vlan10 IP BR vlan20 IP RTR-BR <-> FW-BR	tunnel.1	IP 10.0.1.2	10	Используется
COD	IP COD Network	tunnel.2	IP 10.0.2.2	10	Используется

RTR-BR:

- Создаём туннельные интерфейсы:

```
enable
conf t

interface tunnel.1
description "GRE-to-FW-HQ"
ip address 10.0.1.2/30
ip tunnel 84.212.78.78 63.27.18.18 mode gre
exit

interface tunnel.3
description "GRE-to-RTR-COD"
ip address 10.0.3.2/30
ip tunnel 84.212.78.78 34.95.33.33 mode gre
exit

write memory
```

- Проверить:

```
rtr-br#show interface tunnel.1
Interface tunnel.1 is up
Description: "GRE-to-FW-HQ"
Snmp index: 8
Ethernet address: (port not configured)
MTU: 1476
Tunnel source: 84.212.78.78
Tunnel destination: 63.27.18.18
Tunnel mode: GRE
Tunnel keepalive: disabled
NAT: no
ARP Proxy: disable
ICMP redirects on, unreachable on, ttl-exceeded on
IP URPF is disabled
Label switching is disabled
<UP,BROADCAST,RUNNING,NOARP,MULTICAST>
inet 10.0.1.2/30 broadcast 10.0.1.3/30
Link-local address is fe80::a681:aaff:fe46:36f4/64
total input packets 0, bytes 0
total output packets 0, bytes 0
rtr-br#show interface tunnel.3
Interface tunnel.3 is up
Description: "GRE-to-RTR-COD"
Snmp index: 9
Ethernet address: (port not configured)
MTU: 1476
Tunnel source: 84.212.78.78
Tunnel destination: 34.95.33.33
Tunnel mode: GRE
Tunnel keepalive: disabled
NAT: no
ARP Proxy: disable
ICMP redirects on, unreachable on, ttl-exceeded on
IP URPF is disabled
Label switching is disabled
<UP,BROADCAST,RUNNING,NOARP,MULTICAST>
inet 10.0.3.2/30 broadcast 10.0.3.3/30
Link-local address is fe80::96b5:38ff:fe6f:f8b4/64
total input packets 0, bytes 0
total output packets 0, bytes 0
rtr-br#
```

- Добавляем статические маршруты в сети офиса HQ:

```
conf t

ip route 10.1.1.0/27 10.0.1.1
ip route 10.1.1.32/28 10.0.1.1
ip route 10.1.2.0/24 10.0.1.1

write memory
```

- Проверить:


```
rtr-br#show ip route static
IP Route Table for VRF "default"
S      10.1.1.0/27 [1/0] via 10.0.1.1, tunnel.1
S      10.1.1.32/28 [1/0] via 10.0.1.1, tunnel.1
S      10.1.2.0/24 [1/0] via 10.0.1.1, tunnel.1

Gateway of last resort is not set
rtr-br#
```

- С **ADM-PC** проверить доступность **FW-BR**, **SRV-BR** и **CLI-BR**:

```
[user@adm-hq ~]$ ping -c3 fw-br
PING fw-br.au.team (10.2.0.2) 56(84) bytes of data.
64 bytes from fw-br.au.team (10.2.0.2): icmp_seq=1 ttl=63 time=62.6 ms
64 bytes from fw-br.au.team (10.2.0.2): icmp_seq=2 ttl=63 time=62.8 ms
64 bytes from fw-br.au.team (10.2.0.2): icmp_seq=3 ttl=63 time=61.9 ms

--- fw-br.au.team ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2001ms
rtt min/avg/max/mdev = 61.891/62.415/62.765/0.377 ms
[user@adm-hq ~]$ ping -c3 srv-br
PING srv-br.au.team (10.2.1.10) 56(84) bytes of data.
64 bytes from srv-br.au.team (10.2.1.10): icmp_seq=1 ttl=62 time=51.9 ms
64 bytes from srv-br.au.team (10.2.1.10): icmp_seq=2 ttl=62 time=50.3 ms
64 bytes from srv-br.au.team (10.2.1.10): icmp_seq=3 ttl=62 time=48.9 ms

--- srv-br.au.team ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 48.876/50.371/51.939/1.251 ms
[user@adm-hq ~]$ ping -c3 10.2.2.2
PING 10.2.2.2 (10.2.2.2) 56(84) bytes of data.
64 bytes from 10.2.2.2: icmp_seq=1 ttl=62 time=18.7 ms
64 bytes from 10.2.2.2: icmp_seq=2 ttl=62 time=16.9 ms
64 bytes from 10.2.2.2: icmp_seq=3 ttl=62 time=14.9 ms

--- 10.2.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 14.919/16.821/18.682/1.536 ms
[user@adm-hq ~]$
```

- Настраиваем OSPF для обмена маршрутной информацией с **RTR-COD**:

```
conf t

router ospf 2
passive-interface default
no passive-interface tunnel.3
network 10.0.3.0/30 area 0
exit

write memory
```

- Проверить:

```
rtr-br#show run ospf
!
!
router ospf 1
  passive-interface default
  no passive-interface int1
  network 10.2.0.0/30 area 1
  default-information originate
!
router ospf 2
  passive-interface default
  no passive-interface tunnel.3
  network 10.0.3.0/30 area 0
  no capability cspf
!
rtr-br#
```

RTR-COD:

- Создаём туннельные интерфейсы:

```
enable
conf t

interface tunnel.2
description "GRE-to-FW-HQ"
ip address 10.0.2.2/30
ip tunnel 34.95.33.33 63.27.18.18 mode gre
exit

interface tunnel.3
description "GRE-to-RTR-BR"
ip address 10.0.3.1/30
ip tunnel 34.95.33.33 84.212.78.78 mode gre
exit

write memory
```

- Проверить:

```
rtr-cod#show interface tunnel.2
Interface tunnel.2 is up
  Description: "GRE-to-FW-HQ"
  Snmp index: 7
  Ethernet address: (port not configured)
  MTU: 1476
  Tunnel source: 34.95.33.33
  Tunnel destination: 63.27.18.18
  Tunnel mode: GRE
  Tunnel keepalive: disabled
  NAT: no
  ARP Proxy: disable
  ICMP redirects on, unreachable on, ttl-exceeded on
  IP URPF is disabled
  Label switching is disabled
  <UP,BROADCAST,RUNNING,NOARP,MULTICAST>
  inet 10.0.2.2/30 broadcast 10.0.2.3/30
  Link-local address is fe80::c6ec:9eff:fe6d:69c9/64
  total input packets 0, bytes 0
  total output packets 0, bytes 0
rtr-cod#show interface tunnel.3
Interface tunnel.3 is up
  Description: "GRE-to-RTR-BR"
  Snmp index: 8
  Ethernet address: (port not configured)
  MTU: 1476
  Tunnel source: 34.95.33.33
  Tunnel destination: 84.212.78.78
  Tunnel mode: GRE
  Tunnel keepalive: disabled
  NAT: no
  ARP Proxy: disable
  ICMP redirects on, unreachable on, ttl-exceeded on
  IP URPF is disabled
  Label switching is disabled
  <UP,BROADCAST,RUNNING,NOARP,MULTICAST>
  inet 10.0.3.1/30 broadcast 10.0.3.3/30
  Link-local address is fe80::16e6:dff:fe31:8737/64
  total input packets 0, bytes 0
  total output packets 4, bytes 256
rtr-cod#
```

- Добавляем статические маршруты в сети офиса **HQ**:

```
conf t

ip route 10.1.1.0/27 10.0.2.1
ip route 10.1.1.32/28 10.0.2.1
ip route 10.1.2.0/24 10.0.2.1

write memory
```

- С **ADM-PC** проверить доступность всех устройств **COD**:

```
[user@adm-hq ~]$ ping -c1 rtr-cod
PING rtr-cod.au.team (172.16.1.254) 56(84) bytes of data.
64 bytes from rtr-cod.au.team (172.16.1.254): icmp_seq=1 ttl=63 time=31.0 ms

--- rtr-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 31.009/31.009/31.009/0.000 ms
[user@adm-hq ~]$ ping -c1 sw-cod
PING sw-cod.au.team (172.16.1.0) 56(84) bytes of data.
64 bytes from sw-cod.au.team (172.16.1.0): icmp_seq=1 ttl=63 time=27.8 ms

--- sw-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 27.802/27.802/27.802/0.000 ms
[user@adm-hq ~]$ ping -c1 ha1-cod
PING ha1-cod.au.team (172.16.0.1) 56(84) bytes of data.
64 bytes from ha1-cod.au.team (172.16.0.1): icmp_seq=1 ttl=63 time=32.9 ms

--- ha1-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 32.857/32.857/32.857/0.000 ms
[user@adm-hq ~]$ ping -c1 ha2-cod
PING ha2-cod.au.team (172.16.0.2) 56(84) bytes of data.
64 bytes from ha2-cod.au.team (172.16.0.2): icmp_seq=1 ttl=63 time=59.9 ms

--- ha2-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 59.887/59.887/59.887/0.000 ms
[user@adm-hq ~]$ ping -c1 srv1-cod
PING srv1-cod.au.team (172.16.1.1) 56(84) bytes of data.
64 bytes from srv1-cod.au.team (172.16.1.1): icmp_seq=1 ttl=63 time=17.6 ms

--- srv1-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 17.582/17.582/17.582/0.000 ms
[user@adm-hq ~]$ ping -c1 srv2-cod
PING srv2-cod.au.team (172.16.1.2) 56(84) bytes of data.
64 bytes from srv2-cod.au.team (172.16.1.2): icmp_seq=1 ttl=63 time=17.3 ms

--- srv2-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 17.274/17.274/17.274/0.000 ms
[user@adm-hq ~]$ ping -c1 srv3-cod
PING srv3-cod.au.team (172.16.1.3) 56(84) bytes of data.
64 bytes from srv3-cod.au.team (172.16.1.3): icmp_seq=1 ttl=63 time=22.3 ms

--- srv3-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 22.270/22.270/22.270/0.000 ms
[user@adm-hq ~]$
```

- Настраиваем OSPF для обмена маршрутной информацией с **RTR-BR**:

```
conf t

router ospf 1
passive-interface default
no passive-interface tunnel.3
network 10.0.3.0/30 area 0
network 172.16.0.0/23 area 0
exit

write memory
```

- Добавляем статические маршруты в офис **BR**:

```
ip route 10.2.0.0/30 10.0.3.2
ip route 10.2.1.0/28 10.0.3.2
ip route 10.2.2.0/25 10.0.3.2

write memory
```

Проверяем полную связность между всеми плозадками:

- Из **HQ** в **BR**:

```
[user@adm-hq ~]$ ping -c1 rtr-br
PING rtr-br.au.team (10.2.0.1) 56(84) bytes of data.
64 bytes from rtr-br.au.team (10.2.0.1): icmp_seq=1 ttl=63 time=43.1 ms

--- rtr-br.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 43.108/43.108/43.108/0.000 ms
[user@adm-hq ~]$ ping -c1 fw-br
PING fw-br.au.team (10.2.0.2) 56(84) bytes of data.
64 bytes from fw-br.au.team (10.2.0.2): icmp_seq=1 ttl=63 time=67.7 ms

--- fw-br.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 67.733/67.733/67.733/0.000 ms
[user@adm-hq ~]$ ping -c1 srv-br
PING srv-br.au.team (10.2.1.10) 56(84) bytes of data.
64 bytes from srv-br.au.team (10.2.1.10): icmp_seq=1 ttl=62 time=51.2 ms

--- srv-br.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 51.163/51.163/51.163/0.000 ms
[user@adm-hq ~]$ ping -c1 10.2.2.2
PING 10.2.2.2 (10.2.2.2) 56(84) bytes of data.
64 bytes from 10.2.2.2: icmp_seq=1 ttl=62 time=38.8 ms

--- 10.2.2.2 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 38.849/38.849/38.849/0.000 ms
[user@adm-hq ~]$
```

- Из **HQ** в **COD**:

```

[user@adm-hq ~]$ ping -c1 rtr-cod
PING rtr-cod.au.team (172.16.1.254) 56(84) bytes of data.
64 bytes from rtr-cod.au.team (172.16.1.254): icmp_seq=1 ttl=63 time=11.7 ms

--- rtr-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 11.707/11.707/11.707/0.000 ms
[user@adm-hq ~]$ ping -c1 sw-cod
PING sw-cod.au.team (172.16.1.0) 56(84) bytes of data.
64 bytes from sw-cod.au.team (172.16.1.0): icmp_seq=1 ttl=63 time=68.4 ms

--- sw-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 68.379/68.379/68.379/0.000 ms
[user@adm-hq ~]$ ping -c1 ha1-cod
PING ha1-cod.au.team (172.16.0.1) 56(84) bytes of data.
64 bytes from ha1-cod.au.team (172.16.0.1): icmp_seq=1 ttl=63 time=67.0 ms

--- ha1-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 67.007/67.007/67.007/0.000 ms
[user@adm-hq ~]$ ping -c1 ha2-cod
PING ha2-cod.au.team (172.16.0.2) 56(84) bytes of data.
64 bytes from ha2-cod.au.team (172.16.0.2): icmp_seq=1 ttl=63 time=42.4 ms

--- ha2-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 42.440/42.440/42.440/0.000 ms
[user@adm-hq ~]$ ping -c1 srv1-cod
PING srv1-cod.au.team (172.16.1.1) 56(84) bytes of data.
64 bytes from srv1-cod.au.team (172.16.1.1): icmp_seq=1 ttl=63 time=40.6 ms

--- srv1-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 40.582/40.582/40.582/0.000 ms
[user@adm-hq ~]$ ping -c1 srv2-cod
PING srv2-cod.au.team (172.16.1.2) 56(84) bytes of data.
64 bytes from srv2-cod.au.team (172.16.1.2): icmp_seq=1 ttl=63 time=56.2 ms

--- srv2-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 56.217/56.217/56.217/0.000 ms
[user@adm-hq ~]$ ping -c1 srv3-cod
PING srv3-cod.au.team (172.16.1.3) 56(84) bytes of data.
64 bytes from srv3-cod.au.team (172.16.1.3): icmp_seq=1 ttl=63 time=45.0 ms

--- srv3-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 45.002/45.002/45.002/0.000 ms

```

- Из **BR** в **HQ**:
 - *предварительно необходимо добавить разрешающее правило для любого трафика в FORWARD на FW-HQ*

```
[user@cli-br ~]$ ping -c1 fw-hq
PING fw-hq.au.team (10.1.1.1) 56(84) bytes of data.
64 bytes from fw-hq.au.team (10.1.1.1): icmp_seq=1 ttl=61 time=66.6 ms

--- fw-hq.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 66.590/66.590/66.590/0.000 ms
[user@cli-br ~]$ ping -c1 srv-hq
PING srv-hq.au.team (10.1.1.10) 56(84) bytes of data.
64 bytes from srv-hq.au.team (10.1.1.10): icmp_seq=1 ttl=61 time=14.5 ms

--- srv-hq.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 14.497/14.497/14.497/0.000 ms
[user@cli-br ~]$ ping -c1 10.1.2.128
PING 10.1.2.128 (10.1.2.128) 56(84) bytes of data.
64 bytes from 10.1.2.128: icmp_seq=1 ttl=61 time=52.2 ms

--- 10.1.2.128 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 52.249/52.249/52.249/0.000 ms
[user@cli-br ~]$
```

- И3 **BR Б COD:**

```
[user@cli-br ~]$ ping -c1 rtr-cod
PING rtr-cod.au.team (172.16.1.254) 56(84) bytes of data.
64 bytes from rtr-cod.au.team (172.16.1.254): icmp_seq=1 ttl=62 time=34.0 ms

--- rtr-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 33.967/33.967/33.967/0.000 ms
[user@cli-br ~]$ ping -c1 sw-cod
PING sw-cod.au.team (172.16.1.0) 56(84) bytes of data.
64 bytes from sw-cod.au.team (172.16.1.0): icmp_seq=1 ttl=61 time=71.3 ms

--- sw-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 71.318/71.318/71.318/0.000 ms
[user@cli-br ~]$ ping -c1 ha1-cod
PING ha1-cod.au.team (172.16.0.1) 56(84) bytes of data.
64 bytes from ha1-cod.au.team (172.16.0.1): icmp_seq=1 ttl=61 time=71.1 ms

--- ha1-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 71.091/71.091/71.091/0.000 ms
[user@cli-br ~]$ ping -c1 ha2-cod
PING ha2-cod.au.team (172.16.0.2) 56(84) bytes of data.
64 bytes from ha2-cod.au.team (172.16.0.2): icmp_seq=1 ttl=61 time=87.4 ms

--- ha2-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 87.424/87.424/87.424/0.000 ms
[user@cli-br ~]$ ping -c1 srv1-cod
PING srv1-cod.au.team (172.16.1.1) 56(84) bytes of data.
64 bytes from srv1-cod.au.team (172.16.1.1): icmp_seq=1 ttl=61 time=83.4 ms

--- srv1-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 83.361/83.361/83.361/0.000 ms
[user@cli-br ~]$ ping -c1 srv2-cod
PING srv2-cod.au.team (172.16.1.2) 56(84) bytes of data.
64 bytes from srv2-cod.au.team (172.16.1.2): icmp_seq=1 ttl=61 time=78.0 ms

--- srv2-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 77.981/77.981/77.981/0.000 ms
[user@cli-br ~]$ ping -c1 srv3-cod
PING srv3-cod.au.team (172.16.1.3) 56(84) bytes of data.
64 bytes from srv3-cod.au.team (172.16.1.3): icmp_seq=1 ttl=61 time=99.0 ms

--- srv3-cod.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 99.024/99.024/99.024/0.000 ms
[user@cli-br ~]$
```

- И3 COD Б HQ:


```
[root@sw-cod ~]# ping -c1 fw-hq
PING fw-hq.au.team (10.1.1.1) 56(84) bytes of data.
64 bytes from fw-hq.au.team (10.1.1.1): icmp_seq=1 ttl=62 time=14.8 ms

--- fw-hq.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 14.835/14.835/14.835/0.000 ms
[root@sw-cod ~]# ping -c1 srv-hq
PING srv-hq.au.team (10.1.1.10) 56(84) bytes of data.
64 bytes from srv-hq.au.team (10.1.1.10): icmp_seq=1 ttl=62 time=64.6 ms

--- srv-hq.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 64.578/64.578/64.578/0.000 ms
[root@sw-cod ~]# ping -c1 adm-hq
PING adm-hq.au.team (10.1.1.46) 56(84) bytes of data.
64 bytes from adm-hq.au.team (10.1.1.46): icmp_seq=1 ttl=62 time=15.3 ms

--- adm-hq.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 15.285/15.285/15.285/0.000 ms
[root@sw-cod ~]# ping -c1 10.1.2.128
PING 10.1.2.128 (10.1.2.128) 56(84) bytes of data.
64 bytes from 10.1.2.128: icmp_seq=1 ttl=62 time=53.5 ms

--- 10.1.2.128 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 53.537/53.537/53.537/0.000 ms
[root@sw-cod ~]#
```

- И3 COD Б BR:

```
[root@sw-cod ~]# ping -c1 rtr-br
PING rtr-br.au.team (10.2.0.1) 56(84) bytes of data.
64 bytes from rtr-br.au.team (10.2.0.1): icmp_seq=1 ttl=63 time=74.0 ms

--- rtr-br.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 73.985/73.985/73.985/0.000 ms
[root@sw-cod ~]# ping -c1 fw-br
PING fw-br.au.team (10.2.0.2) 56(84) bytes of data.
64 bytes from fw-br.au.team (10.2.0.2): icmp_seq=1 ttl=62 time=89.5 ms

--- fw-br.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 89.501/89.501/89.501/0.000 ms
[root@sw-cod ~]# ping -c1 srv-br
PING srv-br.au.team (10.2.1.10) 56(84) bytes of data.
64 bytes from srv-br.au.team (10.2.1.10): icmp_seq=1 ttl=61 time=155 ms

--- srv-br.au.team ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 154.820/154.820/154.820/0.000 ms
[root@sw-cod ~]# ping -c1 10.2.2.2
PING 10.2.2.2 (10.2.2.2) 56(84) bytes of data.
64 bytes from 10.2.2.2: icmp_seq=1 ttl=61 time=109 ms

--- 10.2.2.2 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 108.745/108.745/108.745/0.000 ms
[root@sw-cod ~]#
```